Michigan Tech

ODEs as a language of science

Why can prediction fail for deterministic equations?

Allan Struthers

Math Sciences

MA4535 — Nonlinear Dynamics & Chaos | Lecture 1 | September 17, 2026

Outline

Why can prediction fail for deterministic equations?

- Why differential equations matter in science and engineering.
- Three examples: gravity, the Lorenz model, and the Rössler model.
- How computing changed the study of chaos.
- Sprott's simple chaotic equations and circuits [3].

Differential Equations

- If $y(t)$ is position, $v = y'$ is velocity and $a = y''$ is acceleration.
- Ordinary Differential Equations (ODE) relate y to y', y'', ...
- For motion along a line, Newton's law $F = ma$ gives $F = my''(t)$.
  - Starting values $y(0) = y_0$ and $v(0) = v_0$ are Initial Conditions (IC).
- An ODE with IC is an Initial Value Problem (IVP).
- Coupled ODEs form an ODE system.
- Time is always t. We solve for $y(t)$.
  - Scalar $y(t)$, vector $\boldsymbol{y}(t)$, or matrix $\boldsymbol{Y}(t)$.

Second-order ODEs in structures and circuits

Structural vibrations Buildings are modeled as connected masses and springs. Newton's law $F = ma$ gives second-order ODEs.

Electrical circuits Kirchhoff's laws relate voltages around a loop. For charge $q(t)$, component voltages depend on

inductors	$\Delta v \approx q''$
resistors	$\Delta v \approx q'$
capacitors	$\Delta v \approx q$

Kepler's Laws of Planetary Motion

Why can prediction fail for deterministic equations?

1: Elliptical orbits Planets move on ellipses with the Sun at one focus.

2: Equal areas A planet sweeps out equal areas in equal times.

3: Harmonic law The orbital period squared, T^2, is proportional to the semi-major axis cubed, a^3.

Kepler's process: Brahe measured Mars's position without a telescope to roughly 1–2 arcminutes. Circular orbits missed his data by 8 arcminutes; an ellipse fit it. The orbital eccentricities are $\epsilon_{Mars} \approx 0.093$ and $\epsilon_{Earth} \approx 0.018$.

$$r(\theta) = \frac{a(1-\epsilon^2)}{1+\epsilon\cos(\theta)}$$

Newton's gravity model

Why can prediction fail for deterministic equations?

Newton's laws give the motion of n masses m_i at positions $\boldsymbol{r}_i \in \mathbb{R}^3$:

$$m_i \boldsymbol{r}_i'' = -\sum_{\substack{j=1 \\ j \neq i}}^{n} G \frac{m_i m_j (\boldsymbol{r}_i - \boldsymbol{r}_j)}{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^3}, \quad \text{for } i = 1, \ldots, n. \tag{1}$$

Canceling m_i gives this nonlinear ODE system:

$$\boldsymbol{r}_i'' = -G \sum_{\substack{j=1 \\ j \neq i}}^{n} m_j \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^3} \tag{2}$$

Poincaré's study of the three-body problem motivates the tools we will develop [2].

Clockwork Vision

Why can prediction fail for deterministic equations?

The celestial machine is to be likened not to a divine organism but rather to a clockwork. — *Kepler 1605*

An intelligence which, at a given instant, knew all the forces by which nature is animated, and the respective positions of all the beings of which it is composed, and if it were moreover sufficiently vast to subject these data to analysis, would embrace in the same formula the motions of the greatest bodies of the universe and those of the lightest atom. — *Laplace, 1814*

Poincaré and Lorenz

Why can prediction fail for deterministic equations?

One will be struck by the complexity of this figure, which I do not even attempt to draw. Nothing can give us a better idea of the intricacy of the three-body problem, and of all the problems of dynamics in general . . . a kind of lattice-work, a weave, a chain-link network of infinitely fine mesh. — Poincaré 1899

"Predictability: Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?" — Lorenz 1972

If the flap of a butterfly's wings can be instrumental in generating a tornado, it can equally well be instrumental in preventing a tornado. — Lorenz 1972

Lorenz

Why can prediction fail for deterministic equations?

$$\begin{aligned} \dot{x} &= \sigma(y - x) \\ \dot{y} &= x(\rho - z) - y \\ \dot{z} &= xy - \beta z \end{aligned} \qquad \begin{aligned} \sigma &= 10 \\ \rho &= 28 \\ \beta &= \frac{8}{3} \end{aligned}$$

Lorenz found chaos in this simplified weather model. It became the familiar butterfly example of chaos [1].

Rössler

Why can prediction fail for deterministic equations?

$$\dot{x} = -y - z \qquad a = 0.2$$
$$\dot{y} = x + ay \qquad b = 0.2$$
$$\dot{z} = b + z(x - c) \qquad c = 5.7$$

I had no idea what he meant. - *Rössler on being introduced to chaos 1975.*

A prototype equation to the Lorenz model of turbulence contains just one (second-order) nonlinearity in one variable. - *Rössler 1976*

Simple systems can produce complicated behavior. The reason lies in the structure of space and time. Simple 'weaving rules' suffice to determine very complicated motions. - *Rössler 2020*

What changed

- 1890: Poincaré exposed the complexity of three-body motion [2].
- 1960s–70s: Computers made it practical to solve ODEs numerically.
- 1960s–70s: Researchers found chaos in many systems and began to explain it.
- 1990s: More people gained access to ODE solvers.
- Today: Plots make computed trajectories easier to explore.

Rössler Chaos and Hyperchaos

Why can prediction fail for deterministic equations?

1976 Chaos: Std
(a,b,c)=(0.2,0.2,5.7)

$$\dot{x} = -y - z,$$
$$\dot{y} = x + ay,$$
$$\dot{z} = b + z(x - c).$$

$$(\lambda_1, \lambda_2, \lambda_3) = (+, 0, -)$$

1979 Hyperchaos
(a,b,c,d)=(0.25,3,0.5,0.05)

$$\dot{x} = -y - z,$$
$$\dot{y} = x + ay + w,$$
$$\dot{z} = b + xz,$$
$$\dot{w} = -cz + dw.$$

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (+, +, 0, -)$$

The computable **Lyapunov exponents** λ_i will distinguish chaos from hyperchaos.

Sprott jerk ODEs 1–12

Circuit realizations in [3]

No.	Jerk ODE	IC $(y, \dot{y}, \ddot{y})$	Lyapunov exps
1	$\dddot{y} = -2.017\ddot{y} \pm \dot{y}^2 - y$	$(0, 0, \pm 1)$	$0.055,\ 0,\ -2.072$
2 DM	$\dddot{y} = -2.8\dot{y} \pm y + y^2$	$(\mp 0.5, -1, 1)$	$0.002,\ 0,\ -0.002$
3	$\dddot{y} = -0.44\ddot{y} - 2\dot{y} \pm (y^2 - 1)$	$(0, 0, 0)$	$0.105,\ 0,\ -0.545$
4	$\dddot{y} = -0.5\ddot{y} - \dot{y} \pm y \pm y^2$	$(0, \pm 1, 0)$	$0.094,\ 0,\ -0.594$
5	$\dddot{y} = -2\dot{y} \pm (\lvert y \rvert - 1)$	$\pm(-1, -1, 1)$	$0.003,\ 0,\ -0.003$
6	$\dddot{y} = -0.6\ddot{y} - \dot{y} \pm (\lvert y \rvert - 1)$	$(0, 0, 0)$	$0.036,\ 0,\ -0.636$
7 OS	$\dddot{y} = -0.3\ddot{y} - 0.3\dot{y} - D(y) + 1$	$(0, 0, 0)$	$0.042,\ 0,\ -0.342$
8	$\dddot{y} = -0.3\ddot{y} - 0.3\dot{y} - R(y) - 1$	$(0, 0, 0)$	$0.042,\ 0,\ -0.342$
9	$\dddot{y} = -2.9\dot{y} \pm (0.7y - D(y) + 1)$	$\pm(0, -0.5, 0.5)$	$0.005,\ 0,\ -0.005$
10	$\dddot{y} = -2.9\dot{y} \pm (0.7y - R(y) - 1)$	$\pm(0, -0.5, 0.5)$	$0.005,\ 0,\ -0.005$
11 LB	$\dddot{y} = -0.5\ddot{y} - \dot{y} - y + \mathrm{sgn}(y)$	$(0, 0, 0)$	$0.080,\ 0,\ -0.580$
12 AS	$\dddot{y} = -0.5\ddot{y} - \dot{y} + y - \mathrm{sgn}(y)$	$(0, 0, 0)$	$0.080,\ 0,\ -0.580$

Sprott jerk ODEs 13–22

No.	Jerk ODE	IC $(y, \dot{y}, \ddot{y})$	Lyapunov exps
13	$\dddot{y} = -0.7\ddot{y} - \dot{y} - y + H(y)$	$(0, 0, 0)$	$0.035,\ 0,\ -0.735$
14	$\dddot{y} = -0.4\ddot{y} - \dot{y} - y + 2S(y)$	$(0, 0, 0)$	$0.050,\ 0,\ -0.450$
15 AP	$\dddot{y} = -0.4\ddot{y} - \dot{y} + y - 2S(y)$	$(0, 0, 0)$	$0.050,\ 0,\ -0.450$
16 IZ	$\dddot{y} = -0.19\ddot{y} - \dot{y} - y + 2\tanh y$	$(0, 0, 0)$	$0.100,\ 0,\ -0.290$
17	$\dddot{y} = -0.19\ddot{y} - \dot{y} + y - 2\tanh y$	$(0, 0, 0)$	$0.100,\ 0,\ -0.290$
18 AW	$\dddot{y} = -3.7\dot{y} \pm (y - y^3)$	$(0, 0, 0)$	$0.020,\ 0,\ -0.020$
19 BF	$\dddot{y} = -0.6\ddot{y} + 2.8\dot{y} - \dot{y}^3 - y$	$(0, 0, 0)$	$0.110,\ 0,\ -0.710$
20 AR	$\dddot{y} = -0.7\ddot{y} - \dot{y} + y - y^3$	$(0, 0, 0)$	$0.100,\ 0,\ -0.800$
21 EB	$\dddot{y} = -0.35\ddot{y} - \dot{y} - y + y^3$	$(0, 0, 0)$	$0.100,\ 0,\ -0.450$
22 TA	$\dddot{y} = -0.2\ddot{y} - \dot{y} \pm \sin y$	$(0, 0, 0)$	$0.120,\ 0,\ -0.320$

Nonlinearities in Sprott jerk ODEs

$$\text{Diode } D(y) = \begin{cases} y & y \geq 0, \\ 0 & y < 0, \end{cases} = \max(y, 0)$$

$$\text{Relay } R(y) = \begin{cases} 1 & y \geq 0, \\ -1 & y < 0, \end{cases} = \operatorname{sgn}(y)$$

$$\text{Saturation } S(y) = \begin{cases} -1 & y < -1, \\ y & |y| \leq 1, \\ 1 & y > 1, \end{cases} = \max\{-1, \min(y, 1)\}$$

$$\text{Heaviside } H(y) = \begin{cases} 1 & y > 0, \\ 0 & y \leq 0 \end{cases} = (\operatorname{sgn}(y) + 1)/2$$

Summary

- ODEs model change across science and engineering [4].
- Poincaré showed why three-body motion is hard to predict [2].
- Sprott built circuits that exhibit chaos from simple equations [3].

Chaos: When the present determines the future, but the approximate present does not approximately determine the future. — Edward N. Lorenz, 1993 [1]

References

[1] Edward N. Lorenz.
The Essence of Chaos.
University of Washington Press, 1993.

[2] Henri Poincaré.
Sur le problème des trois corps et les èquations de la dynamique.
Acta Mathematica, 13:1–270, 1890.

[3] J. C. Sprott.
Simple chaotic systems and circuits.
American Journal of Physics, 68(8):758–763, 2000.

[4] Allan Struthers and Merle Potter.
Differential Equations: For Scientists and Engineers.
Springer, 2 edition, 2019.